

Suggested Teaching Guidelines for
Next-Gen AI: Deep Neural Networks
PGCP-AI August 2026

Duration: 120 Hours: 40 Theory (T) + 50 Lab (L) + 30 Self-Learning (SL)

Objective: To introduce students with deep neural network concepts.

Prerequisites: Knowledge of fundamentals of programming and basic mathematics & statistical knowledge.

Evaluation method: Theory exam– 40%, Lab exam – 40%, Internal exam – 20%

List of Course material

Text Book:

Deep Learning using Python, by S Lovelyn Rose, L Ashok Kumar, D Karthika Renuka, Wiley India, 1st Ed

Reference Book:

1. Deep Learning, Ian Goodfellow, Yoshua Bengio, Aaron Courville
2. Neural Networks and Learning Machines, Simon Haykin
3. Hands-On Machine Learning with Scikit-Learn and TensorFlow
4. TensorFlow Deep Learning Cookbook
5. Reinforcement Learning with TensorFlow: A Beginner's Guide to Designing Self-learning Systems with TensorFlow and OpenAI Gym Sayon Dutta

Session 1: (2T + 2L + 2SL)

Theory:

Introduction to Deep Neural Network

- Neural Network and its applications, Single layer neural Network
- Activation Functions: Sigmoid, Hyperbolic Tangent, ReLu, Overview of Back propagation of errors

Lab Assignment:

- Implement the different activation functions on the Dataset in the Jupyter notebook.

Sessions 2 & 3: (4T + 4L + 2SL)

Theory:

- Introduction Deep Learning and Neural Networks
- Practical Application of Neural Networks
- "Hello World" of Neural Network (Logistic Regression)
- Cost Function
- Activation
- Gradient Descent for Logistic Regression
- Back propagation

Lab Assignment:

- Implement the main steps of a Shallow Neural Network
 - Understand the dataset
 - Implement your first Forward and Backward propagation
 - Implement activation function, gradient descent

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- Build Neural Network Model
- Test and optimize the model
- Make Predictions

(to be done in Jupyter notebook. Need Jupyter-notebook and libraries installed)

Session 4: (2T + 2L + 2SL)

Theory:

- Introduction to Tensorflow
- Introduction to Pytorch
- Comparison of Tensorflow and Pytorch

Lab Assignment:

- Explore Tensorflow, Pytorch and their libraries.

Session 5: (2T + 4L + 2SL)

Theory:

- Sigmoid Model
- Sigmoid Loss function
- Introduction to Learning Algorithm
- Deriving the Gradient Descent Update Rule
- Probability Interpretation of Sigmoid Output
- Sigmoid Evaluation Metrics

Lab Assignment:

- Plotting Sigmoid 2D
- Standardization using Scikit-Learn
- Train/Test Split
- Implement Sigmoid-Based Binary Classifier
- Plot Loss vs Epochs During Gradient Descent
- Threshold-Based Classification (0.3, 0.5, 0.7)
- Confusion Matrix and Accuracy Evaluation

Session 6: (2T + 4L+2SL)

Theory:

- Shallow Neural Networks
- Hidden Units and Hidden Layers
- Activation function- Tan-h and ReLu
- Forward and Backward propagation with a hidden layer
- Deep Learning notations and Neural Network Representations

Lab Assignment:

- Implement a two-class neural network with a hidden layer
- Implement forward and backward propagation
- Compute the cross-entropy loss

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Sessions 7 & 8: (4T + 4L + 2SL)**Theory:**

- Parameters vs Hyper Parameters
- Regularization
- L1-L2 normalization
- Frobenius norm

Lab Assignment:

- Implement L1-L2 Regularization
- Improve performance of the learning algorithms

Session 9: (2T + 2L + 2SL)**Theory:**

- Dropouts, Early Stopping
- Data Augmentation

Lab Assignment:

- Implement Dropouts, Early Stopping and Data Augmentation.

Session 10: (2T + 2L + 2SL)**Theory:**

- Vanishing Gradient and Exploding Gradient problems
- Gradient Checking

Lab Assignment:

- Implement Gradient problems

Session 11: (2T + 4L)**Theory:**

- Batch Normalization and other methods for data normalization

Lab Assignment:

- Programming exercise for Normalization

Session 12: (2T + 2L)**Theory:**

- Optimization Algorithms, ADAM
- Mini batch gradient descent

Lab Assignment:

- Implement ADAM in python and Tensor flow

Session 13: (2T + 2L + 2SL)**Theory:**

- RMSProp, Momentum and other gradient descent algorithms

Lab Assignment:

- Implement gradient descent algorithms
- Fashion MNIST and digits MNIST exercises using Tensor flow.

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Session 14: (3T + 4L + 3SL)**Theory:**

- Sequential Modelling, Introduction
- Basic building blocks, applications
- Introduction to RNNs
- BERT VS GPT

Lab Assignment:

- Implement RNN in Jupyter Notebook.

Sessions 15 & 16: (4T+4L+3SL)**Theory:****CNN - Convolutional Neural Networks**

- Introduction, basic building blocks, Convolutional Concept
- Inception Network, Transfer Learning and intuition
- Data Augmentation
- Padding, strides, Pooling layers, Fully connected layers
- Introduction to Explainable AI (XAI)
- Grad-CAM for CNN Visualization and Interpretation
- Applications and Use Cases

Lab Assignment:

- Using tensor flow and python implement CNN for ASL (American Sign Language)
- Generate Grad-CAM heatmaps to visualize important image regions used for classification

Session 17: (3T + 5L + 2SL)**Theory:**

- GANs - introductions, basic building blocks
- Implementing GAN
- Diffusion model intuition

Lab Assignment:

- Implement GAN using python and tensor flow.

Session 18: (4T + 5L + 4SL)**Theory:****Tuning Deep Learning Models**

- Deciding Number of Layers, Deciding Number of Neurons
- CNN tuning for better performance
- RNN/LSTM for Time Series Prediction

Trends in Deep Learning

- Echo State Networks / Reservoir Computing
- Auto Encoders, Convolutional Auto Encoder
- Extreme Learning
- Explainable AI using SHAP
- Introduction to Retrieval-Augmented Generation (RAG)
- Vector Databases and Embeddings (Overview)
- Applications of RAG in Enterprise AI Systems

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Deep Learning Case Studies

- Whale Identification (5k Classes)

Lab Assignment:

- Implement CNN using python and tensor flow
- Implement RNN/LSTM using python and tensor flow
- Apply SHAP to explain predictions of a trained neural network model
- Build a simple RAG pipeline using embeddings and vector similarity search on a small document collection